

Surface-water storage in Kansas reservoirs and ponds

Volumes at normal (conservation) pool: the water a reservoir normally holds, excluding flood-control space. Compiled October 5, 2026 for comparison with plant-available soil water.

1. Statewide summary (most defensible values)

Water body class	Count	Storage (acre-ft)	Share	Basis and source
24 federal reservoirs	24	2,673,000	60%	Conservation-pool capacity from bathymetric surveys (1998–2014), projected to 2016. Rahmani et al. (2018), Table 1.
Other reservoirs, 40 ha (99 acres) or larger	115	471,000	11%	Normal storage of every other dam with surface area of 40 ha or more (state fishing lakes, city lakes, power-plant cooling lakes, watershed districts). USACE National Inventory of Dams, data updated Oct 5, 2026.
Small impoundments, under 40 ha (farm ponds and small lakes)	241,295	1,299,000	29%	Statewide pond inventory mapped from NHD, NWI, land cover and aerial imagery; volume from an area–storage regression fitted to NID dams. Callihan (2013).
Total surface-water storage	~241,434	4,443,000	100%	≈ 4.4 million acre-feet (5.5 km³)

Comparison. Plant-available water in the top 24 inches of Kansas soils at field capacity is about 24.3 million acre-feet (30.0 km³), or about **5.5 times** all reservoir and pond storage in the state and about **9 times** the 24 federal reservoirs.

2. The 24 federal reservoirs (conservation pool)

Reservoir	Closed	Original (acre-ft)	Last survey	2016 estimate (acre-ft)	Capacity lost	Survey by
Milford	1967	415,000	2009	366,000	12%	USACE
Wilson	1963	248,000	2008	237,000	4%	KBS
Tuttle Creek	1962	426,000	2009	232,000	45%	USACE
Waconda (Glen Elder)	1967	242,000	2001	209,000	14%	BOR
Perry	1966	243,000	2009	193,000	21%	USACE
Cedar Bluff	1950	186,000	2000	169,000	9%	BOR
Cheney	1964	167,000	1998	156,000	6%	USGS
El Dorado	1979	164,000	2004	156,000	5%	OWRB
Melvern	1972	154,000	2009	148,000	4%	KBS
Clinton	1977	129,000	2009	116,000	10%	KBS
Kirwin	1955	100,000	2013	95,000	5%	KBS
Marion	1967	85,000	2008	79,000	6%	KBS
Hillsdale	1981	82,000	2009	76,000	7%	KBS
Webster	1956	77,000	2011	76,000	1%	KBS
John Redmond	1963	102,000	2014	62,000	40%	KBS
Pomona	1963	71,000	2009	54,000	25%	KBS
Kanopolis	1948	73,000	2007	45,000	39%	KBS
Council Grove	1963	53,000	2008	42,000	19%	KBS

Reservoir	Closed	Original (acre-ft)	Last survey	2016 estimate (acre-ft)	Capacity lost	Survey by
Elk City	1965	53,000	2008	35,000	34%	KBS
Lovewell	1957	45,000	2011	35,000	21%	KBS
Keith Sebelius (Norton)	1964	36,000	2000	34,000	6%	BOR
Big Hill	1980	28,000	2010	23,000	17%	USACE
Fall River	1948	30,000	2010	19,000	35%	KBS
Toronto	1959	28,000	2010	15,000	44%	KBS
Total (24)		3,235,000		2,673,000	17%	

Converted from million m³ in the source (1 million m³ = 810.71 acre-ft); source values are whole million m³, so each row is precise to about ±400 acre-ft. Survey agencies: USACE = U.S. Army Corps of Engineers; BOR = Bureau of Reclamation; KBS = Kansas Biological Survey; USGS = U.S. Geological Survey; OWRB = Oklahoma Water Resources Board.

3. Notes and comments

- **Flood stage (comment, not used in the table).** At the top of their flood-control pools the 24 federal reservoirs can hold about 13.0 million acre-ft, and all 6,483 dams in the NID about 16.0 million acre-ft (NID maximum storage). Adding the small ponds gives roughly 17 million acre-ft. This space is kept empty for floods and is filled only briefly, so it is not a measure of water normally stored. These maximum values are design figures and are not reduced for sediment.
- **No double counting.** Callihan's small-impoundment class (under 40 ha) already includes the roughly 6,300 smaller NID dams (about 385,000 acre-ft), so NID dams under 40 ha are not added separately. The 24 federal reservoirs are taken from Rahmani et al. rather than the NID because the NID lists original design storage (e.g., Tuttle Creek 335,100 acre-ft vs. 232,000 measured) and lists Perry Lake's normal storage as zero.
- **Size cut-off.** 40 ha (99 acres) follows Callihan (2013), who used the American Fisheries Society definition of a small impoundment. Thirty-four NID dams without a surface area were classed as 40 ha or larger when normal storage was at least 500 acre-ft.
- **Sedimentation since 2016.** Applying each reservoir's depletion rate from Rahmani et al. to 2026 lowers the federal total to about 2,566,000 acre-ft (−107,000). The table keeps the published 2016 values because they stay closest to the survey measurements.
- **Uncertainty in the pond estimate.** Pond volumes were not surveyed; they come from a surface area–storage regression. Callihan notes that pond areas mapped in dry years may be low, and that the NID sample favors larger ponds. His text reports 1,299,483 acre-ft and his Table 2 sums to 1,295,285 acre-ft; the difference is negligible.
- **Not included.** Natural wetlands and playas (e.g., Cheyenne Bottoms, Quivira), stream channels, and groundwater. Kansas has very few natural lakes.

4. Sources

- Rahmani, V., Kastens, J.H., deNoyelles, F., Jakubauskas, M.E., Martinko, E.A., Huggins, D.H., et al. (2018). Examining storage capacity loss and sedimentation rate of large reservoirs in the Central U.S. Great Plains. *Water* 10(2), 190. <https://doi.org/10.3390/w10020190>
- U.S. Army Corps of Engineers (2026). National Inventory of Dams, national CSV export, data updated October 5, 2026. <https://nid.sec.usace.army.mil/>
- Callihan, R.A. (2013). Distribution, proliferation and significance of small impoundments in Kansas. M.A. thesis, University of Kansas. <https://hdl.handle.net/1808/12980>
- Kansas Water Office (2022). 2022 Kansas Water Plan (24 federal reservoirs; state storage in 14). <https://www.kwo.ks.gov/water-plan/water-plan>